Lab Assignment 5: Diode Characterization and Applications

Revision February 08, 2023

Summary

In this lab, various diodes (general purpose switching, rectifier, and Zener diodes) are studied for their characteristics in forward and reverse biased modes. These diodes are used in circuit application such as voltage regulator, voltage clipper, peak detector and voltage doublers. The following tasks will be performed:

- Obtain I_D vs V_D characteristics for forward biased and reverse biased diodes using either a curve tracer (TEK571) or Digilent Discovery III.
- Estimate I_S and n from the data obtained in the lab assignment.
- Voltage Regulator using a Zener diode.
- Voltage Clipper.
- Voltage Doubler circuit

Learning Outcomes: After completing this laboratory assignment, you should be able to:

- Obtain diode characteristics and estimate diode equation parameters.
- Simulate diode SPICE characteristics using obtained diode model parameter and compare them.
- Design a voltage regulator circuit using a Zener diode.
- Analyze a voltage limiter/clipping circuit.
- Analyze a voltage doubler circuit and observe the effect of load on its performance in terms of the output voltage and ripple voltage.

Required Equipment

- EE352 Analog parts kit
- Breadboard
- Function Generator
- Oscilloscope
- DMM (2)
- DC Power Supply
- TEK 571 Curve tracer or Discovery III

I. Diode Forward Characteristics:

In this part of the lab assignment, you will obtain the characteristics curves for forward biased diodes (1N4001 and 1N914) and calculate the critical diode parameters such as I_S and n.

Pre-Lab:

(a) A diode equation is given as $I_D = I_S(e^{\frac{V_D}{nV_T}} - 1)$ where 1 being very small compared to $e^{\frac{V_D}{nV_T}}$ term, it can be ignored to derive diode parameters I_S and n. This diode equation can be expressed in a linear form by taking natural log on both sides, i.e. $\ln(I_D) = \ln(I_s) + \frac{V_D}{nV_T}$. Two variables n and I_s can be calculated by taking two sets of (I_D , V_D) data points.

(b) Derive an expression for the dynamic resistance of a diode,

$$r_D = \frac{dv}{di}$$

Plot dynamic diode resistance r_D as a function of diode voltage. Indicate the point of maximum resistance. Is it consistent with the plot of a diode characteristic?

(c) Obtain LTSPICE diode curve by defining a custom diode in a schematic diagram and plotting I_D *vs* V_D. In LTSPICE define a diode model as .model my_diode D(I_s=1E-12, n=1.7) and select V_D as x-axis and I_D as and y-axis . These parameters will be updated later with the calculated values for diodes 1N914 and 1N4001.

Lab Procedures:

1. Using the circuit below obtain 10 data points (I_D , V_D) for diodes IN4001, and 1N914.
2. *Plot I_D vs. V_D for diodes 1N914 and 1N4001 and demonstrate it to the TA*

   Note:
   (i) Make sure DMM1 is configured to measure current, and DMM2 to measure voltage.

   (ii) You will find that there is very small current for voltage less than 0.5V, and at higher voltage the current increases very rapidly. Therefore, be careful while incrementing voltage V.

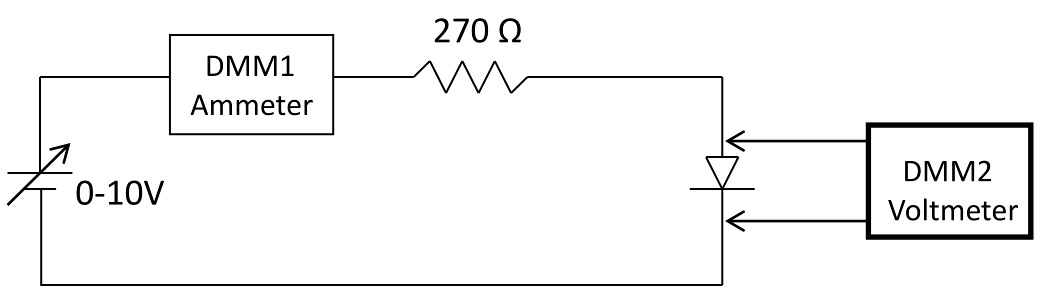

Figure 1. Diode voltage and current measurement

2. Use Curve Tracer TEK571 to obtain the diode characteristics. You may use Digilent Discovery III board if Curve Tracer is not available. Take the picture of plots for both diodes 1N914 and 1N4001. Make sure that you have at least 30mA as maximum currents for these plots.

3. Estimate diode equation variables I_S *and* n using data in part 1.
4. Use these values for I_s and n to plot diode characteristics in LTSPICE. Are these plots similar to the pictures taken in part 2. Comment on the similarity or dissimilarity between the picture taken and LTSPICE plots. Also comment on the I_S and n values for the two diodes used.
5. Determine I_s and n using a least-squares fit of a curve to the data obtained in (1). Use any software available to you for data fitting. Assume room temperature to be 27^{O} C or 300^{O} K to calculate thermal voltage $V_{T.}$

II. Diode Reverse Characteristics

In this lab assignment, the reverse characteristics of a switching diode and a Zener diode are studied.

Pre-lab: None

Lab Procedures:

1. Construct the circuit shown in Figure 2. Make sure that the OP27 is connected to +15V and -15V power supplies and its ground is connected to the ground of the rest of the circuit. Use 1N914 switching diode in the circuit. Measure the amplifier resistors R_1 and R_2 and compute the expected amplifier gain $G_a = 1 + (R_2/R_1)$.

2. You may not have three power supplies (0-10V or higher) available on your bench. You may use function generator output as PS power supply when it is operating with minimum of ac voltage @100 Hz and variable offset voltage as dc power supply. This offset voltage is the value for PS dc output.

3. Reduce PS voltage to zero and measure output voltage, V_{out} at pin 6 of the OP27as shown in Figure 2. This voltage (V_{os1}) is due to op-amp offset voltage and bias currents. You may have to use a 100KΩ, 47KΩ, or 22KΩ resistor for R in the circuit. Recommend starting value of R = 100KΩ. If R is too large, the op-amp can go unstable when the input is 0 volts, due to feedback of the offset currents through R. To check stability, use a x1 scope probe to measure V_{out} with DC coupling on 5 volts/division scale, when the input is set to 0 volts. If there are large oscillations or noise in V_{out} (≥ 1V p-p), reduce R to the next smallest value and re-check. Use the largest value of R that gives you a stable V_{out}, as larger values of R give you larger readings on the DMM and therefore increase your measurement resolution. Make sure you measure and record the resistor R.

4. Vary PS voltage from 0 to 10V (by changing offset voltage of the function generator) in 10 steps and measure and record the op amp output voltage, V_{out}, using a DMM set on volts DC connected between V_{out} and ground. Make sure that the op-amp output voltage is not saturated, i.e. close to (within 1 volt of) power supply voltage for op-amp, when increasing the PS voltage. If the amplifier saturates, use a lower value for R and continue your measurements from the point you left off; note in your lab notebook when you changed the value of R and also what the measured R was. Note that the measurement in step 5 below must also be done for every one of the 10 steps in the measurement of V_{out}.

   The reverse diode current is given as $I_S = (V_{out} - V_{os1}) / (G_a \times R)$.

For each measured V_{out}, compute I_S using the above formula with the values of resistor R and gain G_a that were actually used to take the measurement.

5. For every step on the measurement of V_{out} in part 4, also measure the voltage V_D across the diode, using a second DMM set on volts DC with the positive lead hooked to the cathode side of the diode and the negative lead hooked to the other side of the diode. *Very important:* you cannot measure V_{out} accurately while the second DMM is measuring V_D. You must disconnect *both* leads of the V_D DMM before taking any V_{out} measurement. You may however leave the V_{out} DMM connected throughout the experiment, as it will not affect the V_D measurement.

6. Plot reverse diode characteristic I_S *vs* V_D. Is the value for I_s at the highest value of V_D in agreement with the value calculated earlier? *Demonstrate your results and get it checked by the TA.*

7. Repeat steps 1-6 using Zener diode 1N4735. You might have to reduce resistor R to 10K or even 1K and amplifier gain to a smaller value such as gain of 11 to keep op-amp output voltage V_{out} from saturation. When you are varying PS voltage beyond 4V, you should vary it in smaller increments such as 0.1 or 0.2 volts or so. (Note: once PS gets close to 6.0 V, recommend reducing R to 1K and amplifier gain to 11. Make a note on your lab notebook measurements whenever you switch either R or the amplifier gain, and record measured value of R and value of gain consistent with measured values of the amplifier resistors.)

8. *Demonstrate the Zener diode results to the TA.*

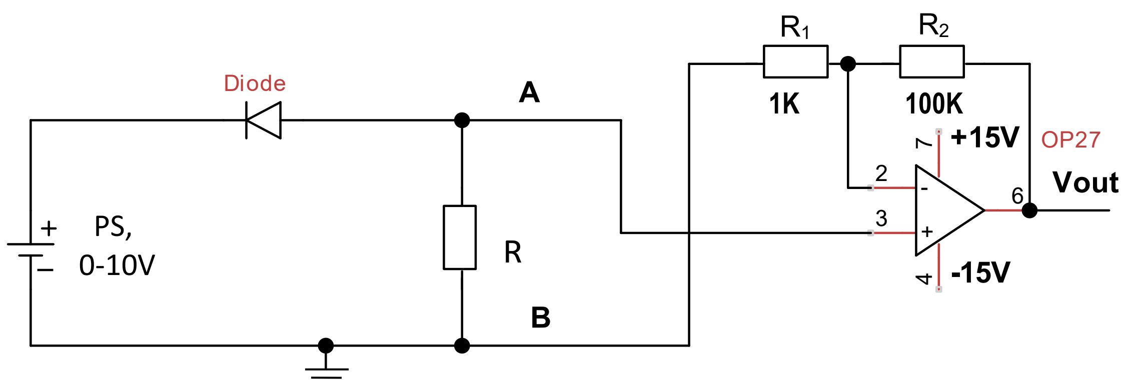

Figure 2. Experimental set up to measure diode reverse characteristics

III. Voltage Regulator

In this lab assignment, we will design a voltage regulator to supply 6.2 V dc voltage. We will use Zener diode 1N4735. Its voltage regulation will be checked at 5mA load.

Pre-lab:

Using the Zener diode (1N4735), design a voltage regulator which can supply current 0-10mA. Note the value of resistor R used. The Zener diode is rated for 6.2V at 1mA. Use 10V power supply for calculations.

Lab Procedures:

1. Construct the circuit as shown in Figure 3. Use R calculated in pre-lab. For 10V power supply use offset voltage of 10V and minimum ac signal from a function generator as was done in part II.

2. Vary load resistance and note dc load voltage and load current for 0 to 10mA in 10 steps. Make sure that R_{load} is never shorted.

3. Plot the load voltage vs. load current; also Zener voltage and Zener current on the same plot. The Zener voltage V_Z = measured load voltage V_L. If V_s is the source voltage and I_L is the measured load current, the Zener current I_Z can be computed from your measured V_Z and I_L as:

$$I_Z = \frac{(V_s - V_Z)}{R} - I_L .$$

4. From the plots estimate the value of Zener diode resistance at 5mA load current.

5. *Demonstrate the working of this voltage regulator to lab TA.*

6. Change ac signal coming from signal generator to 1V $_{p\text{-}p}$ at 1KHz. The output from the function generator is 10V DC + 1V $_{p\text{-}p}$ AC (1KHz) (ripple voltage). Observe the ripple voltage at the load when DC load current is 5mA. Find the ratio of output ripple to input ripple.

7. Observe the output voltage for a load resistance of 200 ohm. Comment on the result.

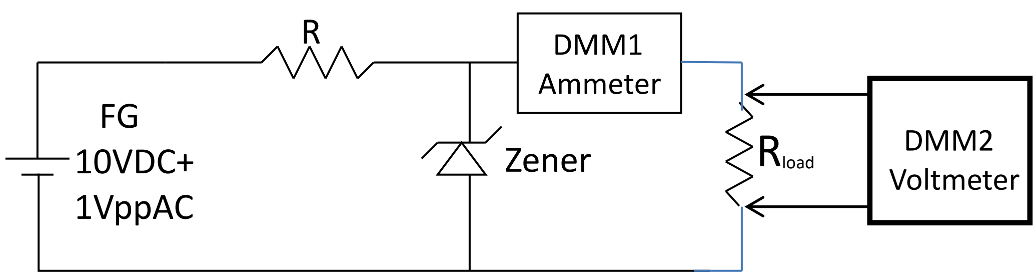

Figure 3. Voltage regulator using Zener diode

IV. <u>Voltage Limiting/Clipping Circuit</u>

In this assignment a voltage limiter circuit is studied for the purpose of limiting/clipping the output voltage.

Pre-Lab:

(a) Plot voltage between nodes A and B as a function of time. The diode has a 0.7V drop when forward biased. The applied voltage, V_{in} at 1KHz is:

(i) $2V_{p\text{-}p}$ sine wave
(ii) $7V_{p\text{-}p}$ sine wave and
(iii) $10V_{p\text{-}p}$ sine wave

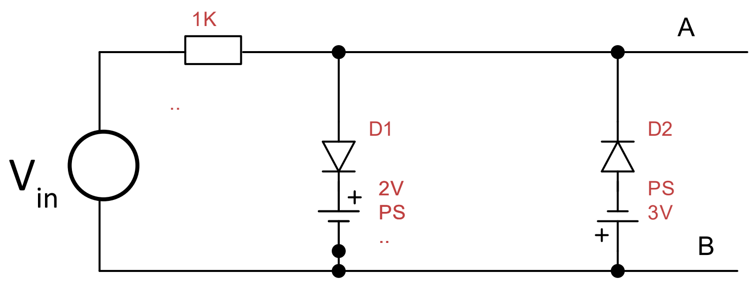

Figure 4. Voltage Limiting/Clipping Circuit

Lab Procedures:

1. Construct the circuit as shown in Figure 4 using switching diodes 1N914. Measure the 1KΩ resistor. Use PS1 at 2V with – terminal grounded and + terminal hooked to D1 cathode, and PS2 at 3V with + terminal grounded and – terminal hooked to D2 anode. This power supply hookup is similar to the power supply hookup for op-amps. The power supply ground should be hooked to the circuit ground (side B) of the circuit in Fig. 4.

2. Apply (i) $2V_{p\text{-}p}$, (ii) $7V_{p\text{-}p}$ and (iii) $10V_{p\text{-}p}$, at 1KHz sine wave input and observe the output across terminals A and B for all three voltage inputs. Take the pictures of the input and output voltages. *Demostrate the circuit operation to TA.*

3. Discuss and compare the observed output waveforms with the expected ones in the pre-lab.

V. <u>Voltage Doubler</u>

In this lab assignment, a voltage doubler circuit is constructed and its output voltage and ripple voltage are studied as output load is changed.

Pre-Lab:

(a) In Figure 5, assume diodes are ideal with 0V drop while conducting. Sketch the output voltage between terminal A and B when no load is connected between terminals A and B; and $10V_{p\text{-}p}$, 1KHz sine wave signal is applied.

Lab Procedures:

1. Construct the circuit as shown in Figure 5. Use diode 1N914.

2. Use C1=C2=1uF, sketch the output voltage for no load (resistor 470Ω and a variable resistor 10KΩ are not connected)

3. Now load the circuit with a 470Ω and a variable resistor 0-10KΩ, and note output voltage V_{out} and ripple voltage V_r (peak to peak) for various values of resistor. V_r should be measured with the oscilloscope on AC coupling, because the DMM will not accurately measure AC voltages at frequencies much greater than 100 Hz. V_{out} can be measured with either a DMM on volts DC, or with the oscilloscope on DC coupling. Take 10 readings for various load resistors. Make sure that the load is never less than 470Ω.

4. Repeat steps 2 and 3 for C1=C2 =10uF.

5. Plot V_{out} and V_r as a function of load resistance. *Demonstrate the working of the circuit to the TA.*

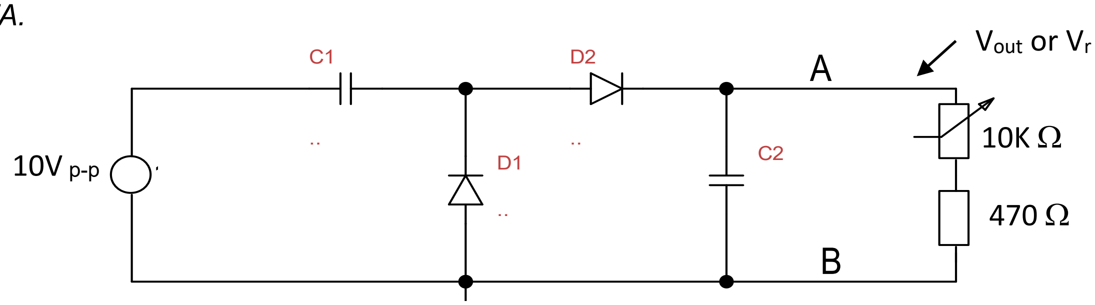

Figure 5. Voltage doubler circuit.

<u>Lab Report:</u>

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet <u>with Laboratory Instructor's initials indicating completed lab demos</u> to your report.

Lab 5 Checklist:

Name:__________________

Lab title and introduction (5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (two or three sentences) explaining the purpose of this lab.

I. Diode Forward Characteristics. (30 pts total)

(a) Switching Diode 1N914 and Rectifier diode 1N4001 (12 pts total)

1. Diagram of circuit (Fig. 1) with measured value for R. (2 pts)
2. Table of measured (I_D, V_D) for switching and rectifier diodes. (5 pts)
3. ________ **DEMO**: Have a TA initial this sheet, indicating he/she has observed your circuit's operation. (5 pts)

(b) Diode Parameter Calculations (18 pts total)

1. Values of Is and n based on two data points (3 pts)
2. Values of Is and n based on least-squares fit of data (5 pts)
3. LTPSPICE I-V curves generated based on Is and n determined from data. (5 pts)
4. Comparison of TEK 571 pictures with LTSPICE I-V curves. (5 pts)

II. Diode Reverse Characteristics (28 pts total)

(a) Diagram of circuit in Figure 2 with the measured value of R. (3 pts)

(b) Switching Diode, 1N914

1. Table of measured values, calculated I_S and plot of data. (5 pts)
2. Compare and discuss I_S obtained in part I and part II. (5 pts)
3. ________ **DEMO**: Have a TA initial this sheet, indicating that he/she has observed your circuit's operation. (5 pts)

(c) Repeat part (b) for Zener diode. Don't compare % difference for I_s.

1. ________ **DEMO**: Have TA initial this sheet, indicating you have obtained data/plot for Zener diode. (5 pts)
2. Discuss switching diode and Zener diode characteristic plots. (5 pts)

III. Zener Diode Voltage Regulator (17 pts total)

1. Draw circuit diagram in Figure 3 with measured value of resistor. (2 pts)
2. Plot load voltage as a function of load current and Zener current. (5 pts)
3. ________ **DEMO**: Obtain TA signature indicating the circuit works. (5 pts)
4. Plot output ripple voltage V_r (p-p) as a function of load current. (5pts)

IV. Diode Clipping/Limiting Circuit (10 pts total)

1. Draw circuit diagram used in Figure 4. (2 pts)
2. Plot input and output voltages for all three voltages. (2 pts)
3. ________ **DEMO:** Obtain TA signature indicating the circuit works. (3 pts)
4. Discussion of circuit's operation (how does it work?). (3 pts)

V. Voltage Doubler Circuit: (10 pts total)

1. Draw circuit diagram (Figure 5). (2 pts)
2. Discussion of circuit's operation. (3 pts)
3. Plot V_{out} and $V_{r(p-p)}$ as a function of load resistor. (2 pts)
4. **DEMO:** Obtain TA signature indicating the circuit works. (3 pts)

